

## \* Merit order Dispatch [MOP]

→ Generation would be taken from that source which delivers the cheapest.

## \* Power System Constraints

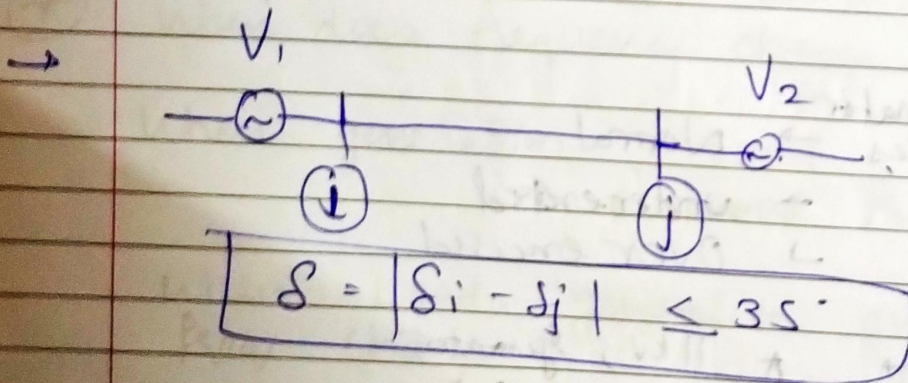
1)  $\Sigma$  Power generated  $P_G = \Sigma P_{\text{Demand}} + \Sigma P_{\text{Losses}}$

2) Inequality  $\rightarrow |V_{\min}| \leq |V_i| \leq |V_{\max}|$

$$P_{Gi\min} \leq P_{Gi} \leq P_{Gi\max}$$

$$Q_{Gi\min} \leq Q_{Gi} \leq Q_{Gi\max}$$

$$f_{\min} \leq f \leq f_{\max}$$



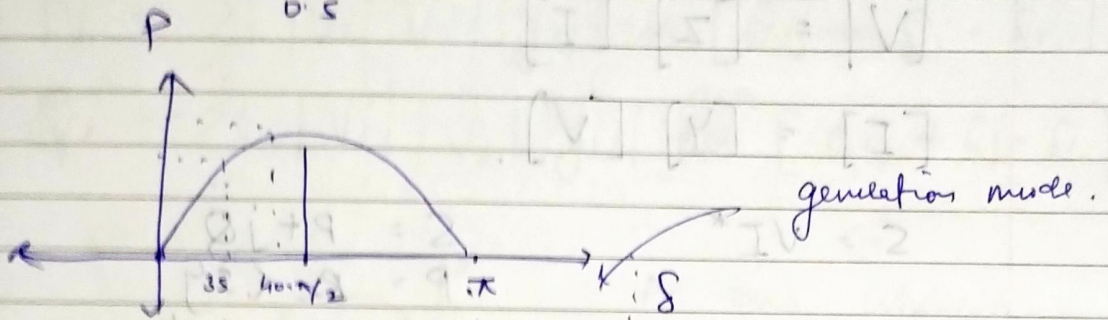
$$P = \frac{|V_i| |V_j|}{X} \sin \delta \cos \theta$$

$$Q = \frac{|V_i| |V_j|}{X} \cos \theta \sin \delta$$



Finding Angle of quantities like Voltage  
 → PMU is used. phasor measurement unit

$$P = \frac{1 \times 1 \sin \delta}{0.5} = 2 \sin \delta$$



In an A.C Network, the behaviour of network is as a Synchronous generator machine.

Beyond  $\pi/2$   $\delta \uparrow$ ,  $P \downarrow$

$\therefore \delta < 35^\circ$   $\delta \in (35, 40)$

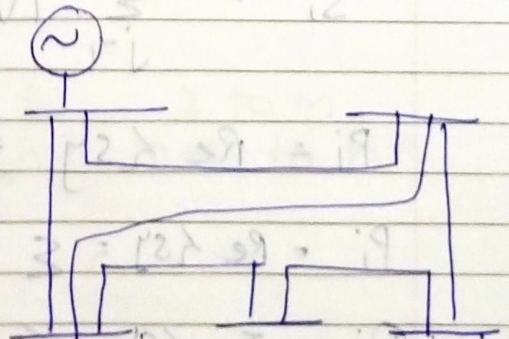
to maintain stability of Power System

→ Reasons to Violate MOD

1. Due to gov policies
2. If line is overloaded
- 3.

→ At every bus these quantities are always monitored

1. voltage magnitude
2.  $\angle \delta_i$
3.  $P_i$
4.  $Q_i$
5.  $S_i$



$$S_i = |V| |I| \angle \theta$$

$$[12 - j2 + j0] \angle 12$$



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$$S = P + jQ \rightarrow \text{absorbing}$$

$$S = P - jQ \rightarrow \text{releasing}$$

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$$[V] = [Z] [I]$$

$$[I] = [Y] [V]$$

$$S = VI^*$$

$$S = P + jQ$$

$$P = \operatorname{Re} \{ S \}$$

$$Q = \operatorname{Im} \{ S \}$$

$$S^* = V^* I$$

$$S_i = V_i^* \sum_{j=1}^n Y_{ij} V_j$$

$$= |V_i| \angle -\delta_i \sum_{j=1}^n |Y_{ij}| |V_j| \angle \delta_j$$

$$= \sum_{j=1}^n |V_i| \angle -\delta_i |V_j| \angle \delta_j \sum_{j=1}^n Y_{ij}$$

$$= \sum_{j=1}^n |V_i| |V_j| \angle (\delta_j - \delta_i)$$

$$S_i^* = \sum_{j=1}^n |V_i| |V_j| \angle (\delta_j - \delta_i)$$

$$P_i = \operatorname{Re} \{ S_i \} = \sum_{j=1}^n |V_i| |V_j| |Y_{ij}| \angle \theta_{ij}$$

$$P_i = \operatorname{Re} \{ S_i \} = \sum_{j=1}^n |V_i| |V_j| |Y_{ij}| \cos [\theta_{ij} + \delta_i - \delta_j]$$

$$Q_i = \operatorname{Im} \{ S_i \} = \sum_{j=1}^n |V_i| |V_j| |Y_{ij}| \sin [\theta_{ij} + \delta_i - \delta_j]$$



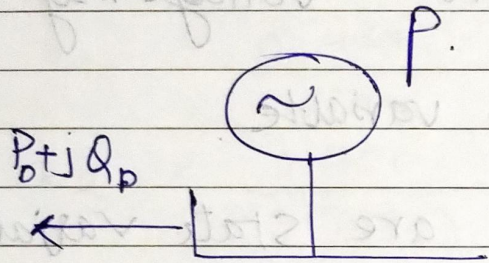
# Static Load flow Eqn

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$$P_i = \sum_{j=1}^n |V_i| |V_j| |y_{ij}| \cos [\theta_{ij} + \delta_j - \delta_i]$$

$$Q_i = - \sum_{j=1}^n |V_i| |V_j| |y_{ij}| \sin [\theta_{ij} + \delta_j - \delta_i]$$



A bus wherein voltage (terminal) and Power (active) is known as PV Bus & generator bus.

A bus where  $P, Q$  are known, are PQ Bus, Load Bus.

Any voltage control bus is PV Bus. ( $P=0$ )

→ Slack / Swing / Reference Bus.

A generator would supply losses to the bus that bus is slack bus.  
generator → highest rating bus.

Known →  $V, \angle \delta = 0$

Unknown →  $P \& Q$

→ slack bus → 1

PV → 2 to m

2 to m

PQ → m+1 to n

m+1 to n



# Gauss Siedel, Newton Raphson.

1. Use iterative Methods

2. Initialize  $V$ ,  $\angle \delta$   
 $V = 1 \text{ pu}$   $\angle \delta = 0^\circ$

$(n-1) \rightarrow$  unknown.

$(n-m) \rightarrow$  unknown voltage mag

Voltage  $\rightarrow$  state variable

Voltage mag,  $\angle \delta$  are state variables



① Gauss Seidel Method {Fixed point method}

② Newton Raphson

Q]  $x \sin x = 0.451$

→  $x^{(0)} = 0.6$

$$x_1 = \frac{0.451}{\sin x} = \frac{0.451}{\sin(0.6)} = 0.7987$$

$$x_2 = \frac{0.451}{\sin(0.7987)} = 0.6294$$

$$x_3 = \frac{0.451}{\sin(0.6294)} = 0.7661$$

$$x_4 = \frac{0.451}{\sin(0.7661)} = 0.6504$$

$$x_5 = 0.7448$$

$$x_6 = 0.6653$$

$$x_7 = 0.7306$$

$$\text{Error}_{12} = \left| \frac{0.6294 - 0.7987}{0.7987} \right| \times 100 = 0.1987$$

$$\text{Error } x_8 = 0.6758 \quad x_{14} = 0.6919$$

$$x_9 = 0.7209 \quad x_{15} = 0.7068$$

$$x_{10} = 0.6832 \quad x_{16} = 0.6944$$

$$x_{11} = 0.7144 \quad x_{17} = 0.7047$$

$$x_{12} = 0.6883 \quad x_{18} = 0.6919$$

$$x_{13} = 0.7099 \quad x_{19} = 0.7068$$



Solving by Newton Raphson,

$$x \sin x = 0.451$$

$$f(x) = x \sin x - 0.451 =$$

$$f'(x) = x(\cos x) + (1) \sin x$$

$$f'(x) = x \cos x + \sin x$$

$$f(0.6) = \left[ 0.6 \sin(0.6) - 0.451 \right] \\ - \frac{(-0.1122)}{+0.0694} = 0.7059$$

$$= 0.7059$$

$$f(0.7) = 0.7059$$

$$\rightarrow \text{error} = 0.7059 - 0.6 = 0.1059 \times 100$$

$$f(0.7059) = 0.7059 + \frac{f(x)}{f'(x)} \\ = 0.7059 + \frac{0.0069}{0.0069} \\ = 0.7181$$

$$= 0.6931$$

$$x_3 = 0.6931 - \frac{(-8.16 \times 10^{-3})}{1.172}$$

$$x_3 = 0.7000$$



# Gauss Seidel Method

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case - All busses are PQ busses  
except slack bus (PV)

↓

P, Q are unknown

|V|,  $\delta \rightarrow$  are known.

n bus system

2n eqn

$2(n-1) \rightarrow |V|, \delta$

$$V_i = \frac{1}{y_{ii}} \left[ \frac{P - jQ}{V_i^*} - \sum_{j=1}^n y_{ij} V_j \right]$$

flat voltage start

→  $|V_i| = 1 \text{ p.u.}$  (voltages at bus = 1 p.u.)  
 $\delta_i = 0$

- ① Calculate  $y_{bus}$
- ② Find injection, flat voltage
- ③ Initial guess
- ④ solve.

$$V^{(k+1)} = V^{(k)} + \alpha (V^{(k+1)} - V^{(k)})$$

$$\alpha = 1.6$$

$$|V^{(k+1)}| - |V^{(k)}| \leq \epsilon \leftarrow \text{threshold value.}$$



Sources of Reactive Power :- Generator,  
Transformer, Capacitor Bank.

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## \* Generator Reactive Capability Limits

$$P_{Gmin} \leq P_G \leq P_{Gmax}$$

$$Q_{Gmin} \leq Q_G \leq Q_{Gmax}$$

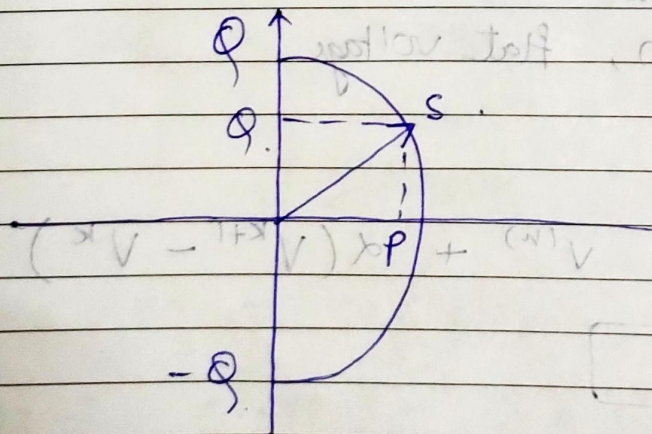
Generator  $\begin{cases} \rightarrow \text{Normal mode} \\ \rightarrow \text{Overexcitation mode} \\ \rightarrow \text{Under excitation mode} \end{cases}$

## \* Voltage drop occurs in transmission line due to reactive power

## \* Armature Heat limit

$$S = P + jQ$$

$$S^2 = P^2 + Q^2$$



$$|S| = \sqrt{P^2 + Q^2} = \sqrt{(1-x)^2 V^2 + (1+x)^2 V^2}$$

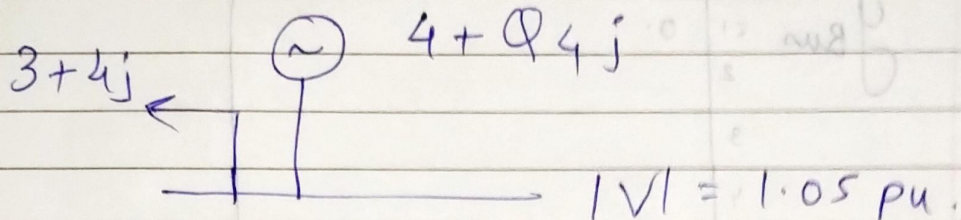
Field Heat limit

End Region Heating



eg

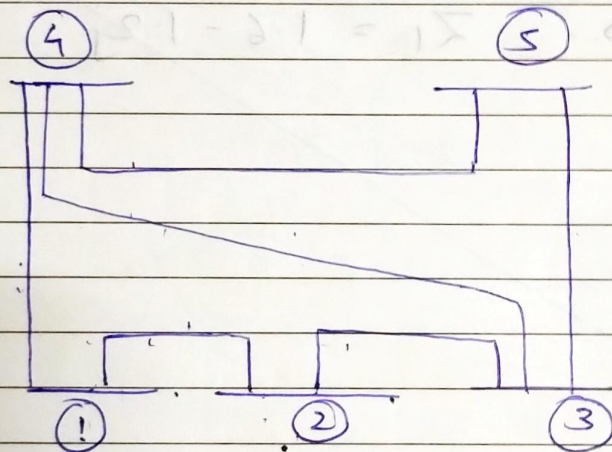
For a PV bus



Reactive power (S) = ?

$$S = (4 - 3) + (0.4 - 4)j$$

Q]



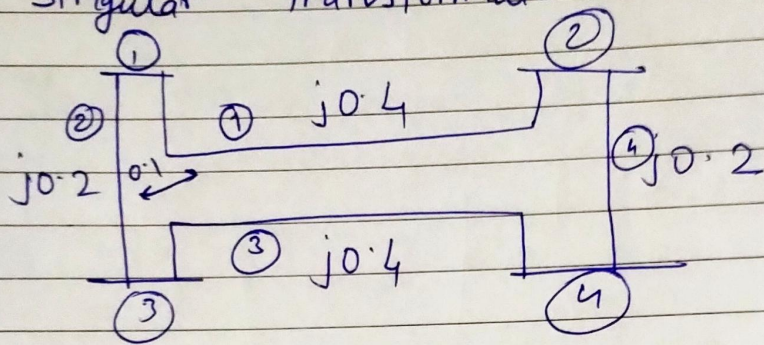
$$(0.02 + j0.2) \text{ pu}$$

Case 1 All buses are PQ buses except slack bus.

| Slack | Bus No. | Load |     | Generation |    | Voltage |
|-------|---------|------|-----|------------|----|---------|
|       |         | P    | Q   | P          | Q  |         |
| Slack | 1       | NS   | NS  | NS         | NS | 1.210   |
| PQ    | 2       | 0.4  | 0.3 | 0          | 0  | NS      |
| PQ    | 3       | 0.6  | 0.3 | 0          | 0  | NS      |
| PQ    | 4       | 0.6  | 0.3 | 0          | 0  | NS      |
| PQ    | 5       | 0.5  | 0.2 | 0          | 0  | NS      |
|       | 6       |      |     |            |    |         |



## Singular Transformation Method



$$Z_{\text{primitive}} = \begin{bmatrix} j0.2 & j0.1 & 0 & 0 \\ j0.1 & j0.4 & 0 & 0 \\ 0 & 0 & 0.2j & 0 \\ 0 & 0 & 0 & 0.4j \end{bmatrix}$$

$$Y_{\text{primitive}} = \text{Inv}(Z_p)$$

making bus incidence matrix

$$\Rightarrow A = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

finding  $Y_{\text{bus}}$

$$Y_{\text{bus}} = [A]^T [Y_{\text{primitive}}] [A]$$

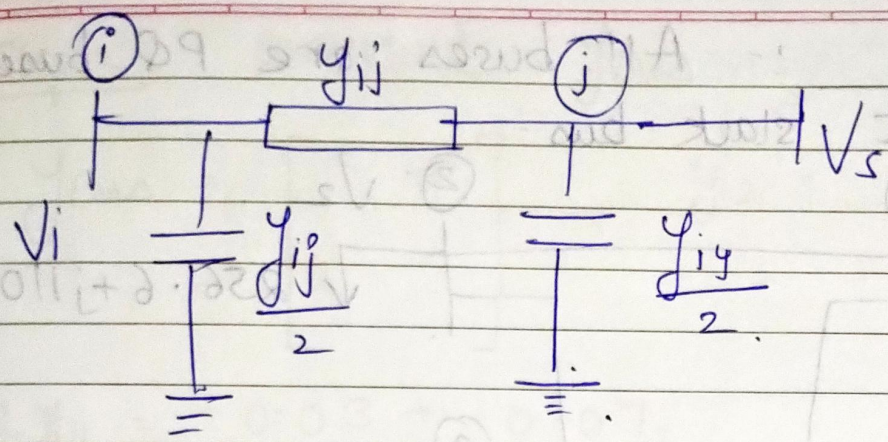
$$= \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} -5.7j & 1.42j & 0 & 0 \\ 1.42j & -2.85j & 0 & 0 \\ 0 & 0 & -5j & 0 \\ 0 & 0 & 0 & -2.5j \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$



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Q]



$$I_{ij} = (V_i - V_j) y_{ij} + V_i \frac{y_{ij}}{2}$$

$$S_{ij} = V_{ij} I_{ij}^*$$

$$I_{ji} = (V_i - V_j) y_{ij} + V_j \frac{y_{ij}}{2}$$

$$S_{ij} = V_j I_{ij}^*$$

$$S_{\text{Line loss}} = S_s - S_R$$

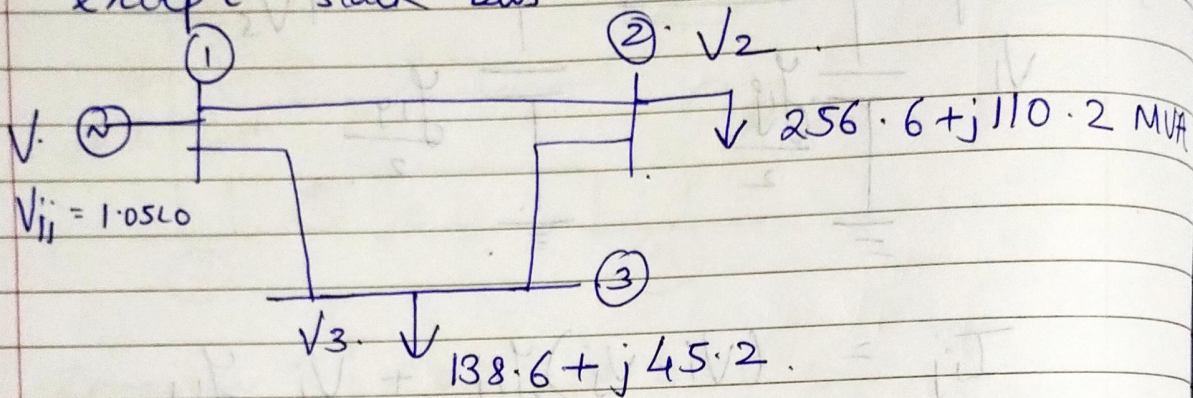
$$= S_{ij} - (-S_{ji})$$

$$S_{\text{Total}} = \sum_{i=1}^n S_{\text{Loss}}$$



Q]

Case 1 :- All buses are PQ buses except slack bus.



1. Find voltages at all buses.
2. Find slack bus and reactive power, real power.

| Line | Z                     |
|------|-----------------------|
| 1-2  | $0.02 + j0.04$ pu.    |
| 1-3  | $0.01 + j0.03$ pu.    |
| 2-3  | $0.0125 + j0.025$ pu. |

→ Take base MVA = 100 MVA

Step 1 :- Convert load in per unit

$$SD_1 = \frac{256.6 + j110.2}{100} = 2.56 + j1.102 \text{ pu}$$

$$SD_2 = \frac{138.6 + j45.2}{100} = 1.38 + j0.452 \text{ pu}$$



Step 2 construct  $Y_{Bus}$

$$Y_{Bus} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \quad Y_{12} = 10 - 20j$$

$$Z_{11} = 0.03 + 0.07j$$

$$Y_{11} = \frac{1}{Z_{11}} = \frac{1}{0.03 + 0.07j} = 52.1724 - j12.068$$

$$Y_{Bus} = \begin{bmatrix} 52.1724 - j12.068 & -10 + 20j & -10 + 30j \\ -10 + 20j & 26 - 62j & -16 + 32j \\ -10 + 30j & -16 + 32j & 26 - 62j \end{bmatrix}$$

Step 3 :- Bus Injection

$$S_2 = P_2 + jQ_2 = -2.56 - j1.102 \text{ pu}$$

$$S_3 = P_3 + jQ_3 = -1.38 - j0.452 \text{ pu}$$

Step 4 :- Initial guess considering flat voltage start

$$|V_2^{(0)}| = |V_3^{(0)}| = 1 \text{ pu}$$

$$\delta_2^{(0)} = \delta_3^{(0)} = 0$$



First Iteration

$$V_i = \frac{1}{y_{ii}} \left[ \frac{P - jQ}{V_i^*} - \sum_{\substack{j=1 \\ j \neq i}}^n y_{ij} V_j \right]$$

$$V_2^{(1)} = \frac{1}{y_{22}} \left[ \frac{P_2 - jQ_2}{V_2^*} - y_{21} V_1^{(0)} - y_{23} V_3^{(0)} \right]$$

$$V_2^{(1)} = \frac{1}{(26 - j52)} \left[ \frac{-2.56 - j1.102}{1} \right]$$

$$- \left\{ \begin{array}{l} (-10 + 20j) \times 1.05 \angle 0^\circ \\ (-16 + 32j) \times 1 \angle 0^\circ \end{array} \right\}$$

$$= (0.0076 + 0.0153) \times$$

$$- \left\{ (-10.5 + 21j) - (-16 + 32j) \right\}$$

$$= (0.0076 + 0.0153) - [5.5 - 11j]$$

$$= 0.12595 + 0.2519j$$

$$V_2^{(1)} = 0.98302 \angle -1.807^\circ$$



$$V_3^{(1)} = \frac{1}{Y_{33}} \left\{ \frac{P_3 - jQ_3}{V_3^*} - Y_{31} V_1^{(0)} - Y_{32} V_2 \right\}$$

$$= \frac{1}{26 - 62j} \left\{ \frac{(-1.386 + j(0.452))}{1} \right.$$

$$- (-10 + j30 \times 1.05 \angle 0)$$

$$- (-16 + 32j \times 0.9802$$

$$\angle -1.801)$$

$$= 0.0057 + 0.0137j$$

$$\left\{ \begin{aligned} &(-1.383 + 0.452j) - (-10 + 31.5j) \\ &- (34.2638 + 7.93j) \end{aligned} \right.$$

$$= -25.6468 - 39.008j$$

$$V_3 = \frac{-25.6468 - 39.008j}{1.00172 \angle -2.0172}$$

$$\text{error} = \begin{bmatrix} |V_2^{(1)} - V_2^{(0)}| \\ |V_3^{(1)} - V_3^{(0)}| \end{bmatrix}$$

$$= \begin{bmatrix} 0.01698 \\ 0.00172 \end{bmatrix}$$

Solve until  $\epsilon = 0.00001$

$$V_2^{(1)} = V_2^{(0)} + \alpha (V_2^{(1)} - V_2^{(0)})$$

$$\alpha = 1.6$$



$$S_1 = V_1 I_1^*$$

$$S_1^* = V_1^* I_1 = V_1^* \sum_{j=1}^n y_{ij} V_j$$

$$= V_1^* [y_{11} V_1 + y_{12} V_2 + y_{13} V_3]$$

$$= 1.05 [20 - 50j \times 1.05$$

$$+ (-10 + 20j \times 0.98 - j0.05$$

$$+ (-10 + 30j \times 1 - j0.05)]$$

$$= 1.05 [21 - 52.5j + (-8.8 + 20.1j) + (-8.5 + 30.5j)]$$

$$= 3.885 - 1.995j$$

$$= 3.885 - 1.995j$$

$$S_1 = 4.3672 \angle -1.27$$

$$S_1^* = P - jQ$$

$$P = 388 \text{ MW}$$

$$Q = 127 \text{ MW}$$



$$y_{12} = -y_{12}$$

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\* Line flow and Line losses

$$\begin{aligned} I_{12} &= y_{12} (V_1 - V_2) \\ &= (10 - 20j) (1.05 - j0 - 1.048 - j0.06) \\ &= 1.29 - 0.8j \end{aligned}$$

$$I_{12} = 2.0615 \angle -22.83^\circ$$

$$I_{12} = 2.0615 \angle -22.83^\circ$$

$$I_{21} = -I_{12} = 2.0615 \angle 157.166^\circ$$

$$\begin{aligned} I_{23} &= y_{23} (V_2 - V_3) \\ &= (10 - j30) [0.98 - j0.06 - (1.00172 \angle -2.017^\circ)] \end{aligned}$$

$$I_{13} = 20 \angle j1.0$$

$$I_{31} = -20 + j1.0$$

$$I_{32} = -I_{23}$$

$$\begin{aligned} I_{23} &= y_{23} [V_2 - V_3] \\ &= (16 - 32j) [0.98302 \angle -1.807^\circ - (1.00172 \angle -2.0172^\circ)] \\ &= -0.1606 + 0.662j \end{aligned}$$

$$I_{23} = 0.6812 \angle 103.63^\circ$$

$$I_{23} = -0.64 + j0.48$$

$$I_{32} = +0.64 - j0.48$$

My

Man



# Line Losses

$$\rightarrow S_{42} = S_{12} + S_{21} = 8.5 + j17.0 \text{ MVA}$$

$$\rightarrow S_{L1,3} = S_{13} + S_{31} = 5.00 + j15.0 \text{ MVA}$$

$$\rightarrow S_{23} = S_{23} + S_{32} = 0.8 + j1.60 \text{ MVA}$$

$$S_{12} = V_1 I_{12}^* = 1.05 \times (1.9 + j0.8) = 1.99.5 + j84 \text{ MVA}$$

$$S_{21} = V_2 I_{21}^* = (0.98 - j0.06) \times (-1.9 - j0.8) = -1.91 - j67 \text{ MVA}$$

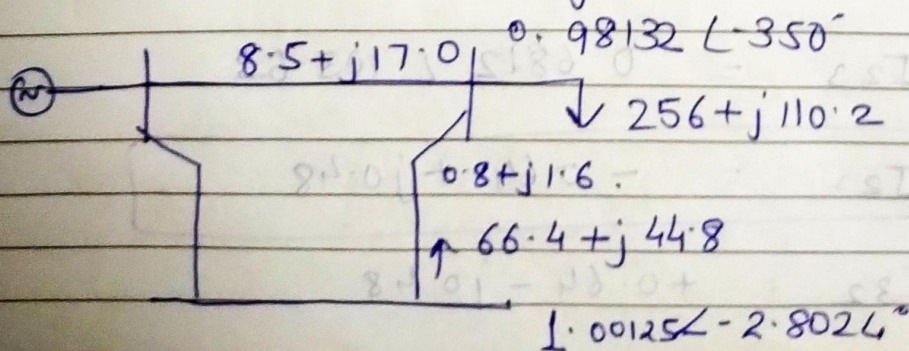
$$S_{13} = V_1 I_{13}^* = 210 + j105 \text{ MVA}$$

$$S_{31} = -205 - j90 \text{ MVA}$$

$$S_{23} = V_2 I_{23}^* = -65.6 - j43.2 \text{ MVA}$$

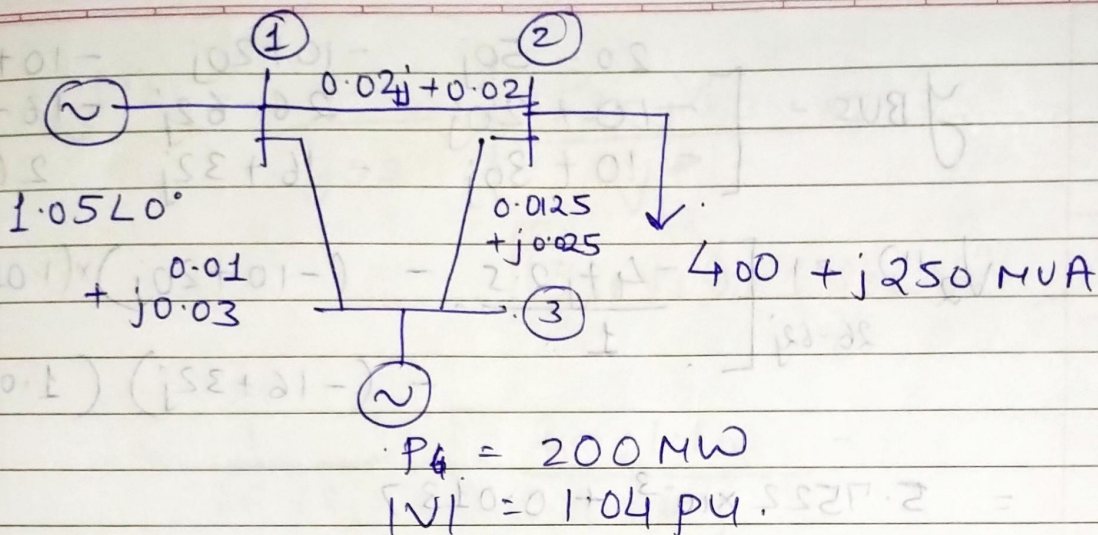
$$S_{32} = V_3 I_{32}^* = 66.4 + j44.8 \text{ MVA}$$

⊛ Load | Power flow diagram





25/08/25



- ① Step 1 - Convert load in pu values  
Base MVA = 100 MVA

$$P_{GD2} = \frac{400 + j250}{100} = 4 + j2.5$$

$$P_{G2} = 2 \text{ pu} \quad S_3 = 2.0 + j0.3$$

$$S_2 = -4 - j2.5$$

- ② Unknown values :-  
 $|V_2|$ ,  $\angle \delta_2$ ,  $\angle \delta_3$ ,  $Q_3$

- ③ Initial guesses:  
Flat voltage start

$$V_2^{(0)} = 1 + j0$$

$$\angle \delta_3^{(0)} = 0$$

$$V_3^{(0)} = 1.04 + j0$$

$$V_2^{(1)} = \frac{1}{y_{22}} \left[ \frac{P_2 - jQ_2}{V_2^*} - \sum y_{21} V_1^0 - y_{23} V_3^0 \right]$$



$$Y_{BUS} = \begin{bmatrix} 20 - 50j & -10 + 20j & -10 + 30j \\ -10 + 20j & 26 - 62j & -16 + 32j \\ -10 + 30j & -16 + 32j & 26 - 62j \end{bmatrix}$$

$$V_2^{(1)} = \frac{1}{26 - 62j} \left[ \frac{-4 + j2.5}{1} - (-10 + 20j)(1.05 \angle 0) - (-16 + 32j)(1.04 \angle 0) \right]$$

$$= 5.7522 \times 10^{-3} + j0.0137$$

$$\left[ -4 + j2.5 - [-10.5 + 21j] - [-16.64 + 33.28j] \right]$$

$$= 0.97462 - j0.042307$$

$$V_2^{(1)} = 0.97462 - j0.042307$$

For PV buses, find Q.

$$Q_3 = -\text{Im} \{ V_3^* I_3 \}$$

$$Q_3 = -\text{Im} \{ V_3^* \sum y_{ij} V_j \}$$

$$Q_3^{(1)} = -\text{Im} \left\{ V_3^{(0)*} \left[ y_{31} V_1^{(1)} + y_{32} V_2^{(1)} + y_{33} V_3 \right] \right\}$$

$$= -41.7039 + j101.6238$$

$$= \text{Im} \{ 2.39 - j1.598 \}$$

$$Q_3^{(1)} = -j1.598 \approx \underline{\underline{1.16 \text{ pu}}}$$



$$V_3^{(1)} = \frac{1}{y_{33}} \left[ \frac{P_3 - jQ_3}{V_3^*} - [(y_{31} V_1) - (y_{32} V_2^{(1)})] \right]$$

$$= \frac{1}{26.62j} \left[ \frac{2 - j1.16}{(1.04)} - \right]$$

$$\left[ \begin{aligned} &(-10 + 30j \times 1.05) \\ &- (-16 + 32j \times (0.9746 - j0.04)) \end{aligned} \right]$$

$$= \frac{1}{26.62} \left[ \left( \frac{2 - j1.16}{1.04} \right) - \left[ \begin{aligned} &(-10 + 30j \times 1.05) \\ &- [14.3136 + 31.8272j] \end{aligned} \right] \right]$$

$$V_3^{(1)} = 1.038 - j0.005170$$

$$\angle \delta_3^{(1)} = \angle (1.038 \angle -0.2853)$$

$$\angle \delta_3 = -0.2853$$

$$V_3^{(1)} = 1.0384 \angle -0.2853$$



# Newton Raphson.

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- ▶ A matrix containing elements as 1st derivative is called as Jacobian.
- Jacobian gives information about slope
- whereas a Hessian Matrix has 2nd derivative elements and tells us the curvature.

## \* Static Load Flow Equation

$$\begin{aligned} \rightarrow \quad Q_{iz} &= \sum_{j=1}^n |V_i| |V_j| |Y_{ij}| \sin(\theta_{ij} + \delta_j - \delta_i) \\ P_i &= \sum_{j=1}^n |V_i| |V_j| |Y_{ij}| \cos(\theta_{ij} + \delta_j - \delta_i) \end{aligned}$$

## → Jacobian Method

$$[J] [\Delta x] = -[f]$$

$$[\Delta x] = -\frac{[f]}{[J]}$$

$$\Delta x^{(i)} = -[J^{(i)}]^{-1} [f^{(i)}]$$

$$[x^{(i)}] = [x^{(0)}] + [\Delta x^{(i)}]$$



Use NR at PQ

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$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix} = \begin{bmatrix} f_1(x_1, \dots, x_n) \\ f_2(x_1, \dots, x_n) \\ \vdots \\ f_n(x_1, \dots, x_n) \end{bmatrix}$$

$$\rightarrow P_i = \operatorname{Re} \left\{ V_i^* \left( \sum y_{ik} V_k \right) \right\}$$

$$Q_i = -\operatorname{Im} \left\{ V_i^* \left( \sum y_{ik} V_k \right) \right\}$$

$$f_{ip} = (|V|, \delta) = P_i(\text{specified}) - P_i(\text{calculated}) = 0$$

$$f_{iq} = (|V|, \delta) =$$

Residual Vector

$$\begin{aligned} f_{ip} &= P_i(|V|, \delta) = P_i(\text{specified}) - P_i(\text{calculated}) = \Delta P_i \\ f_{iq} &= Q_i(|V|, \delta) = Q_i(\text{specified}) - Q_i(\text{calculated}) = \Delta Q_i \end{aligned}$$



Case I :- All buses are PQ buses

Bus

except slack bus

|          |                                     |                                          |                                     |                                          |     |                                     |                                          |              |                   |
|----------|-------------------------------------|------------------------------------------|-------------------------------------|------------------------------------------|-----|-------------------------------------|------------------------------------------|--------------|-------------------|
| 2        | $\frac{\partial P_2}{\partial V_2}$ | $\frac{\partial P_2}{\partial \delta_2}$ | $\frac{\partial P_2}{\partial V_3}$ | $\frac{\partial P_2}{\partial \delta_3}$ | ... | $\frac{\partial P_2}{\partial V_n}$ | $\frac{\partial P_2}{\partial \delta_n}$ | $\Delta V_2$ | $\Delta \delta_2$ |
| $\vdots$ |                                     |                                          |                                     |                                          |     |                                     |                                          |              |                   |
| $\vdots$ | $\frac{\partial Q_2}{\partial V_2}$ | $\frac{\partial Q_2}{\partial \delta_2}$ | $\frac{\partial Q_2}{\partial V_3}$ | $\frac{\partial Q_2}{\partial \delta_3}$ | ... | $\frac{\partial Q_2}{\partial V_n}$ | $\frac{\partial Q_2}{\partial \delta_n}$ | $\Delta V_3$ | $\Delta \delta_3$ |
| 3        | $\frac{\partial P_3}{\partial V_2}$ |                                          |                                     |                                          |     |                                     |                                          |              |                   |
| $\vdots$ |                                     |                                          |                                     |                                          |     |                                     |                                          |              |                   |
| n        |                                     |                                          |                                     |                                          |     |                                     |                                          |              |                   |
|          | 2                                   |                                          | 3                                   |                                          | ... |                                     | n                                        |              |                   |

$$N_{im} = \frac{\partial P_i}{\partial V_m}$$

$$H_{im} = \frac{\partial P_i}{\partial \delta_m}$$

$$J_{im} = \frac{\partial Q_i}{\partial V_m}$$

$$L_{im} = \frac{\partial Q_i}{\partial \delta_m}$$

$$J_{im} \quad L_{im}$$



Jacobian is as sparse as  $Y_{bus}$ .

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PV Bus  $\rightarrow P, |V|$  constant

$$f_{ip}(|V|, \delta) = P_{\text{specified}} - P_{\text{calculated}} = \Delta P.$$

Say,  $i$ th bus is PV bus and  $m$ th bus is PQ bus.

$$\begin{bmatrix} V_m & \delta_m \\ \vdots & \vdots \\ \frac{\partial P_i}{\partial V_m} & \frac{\partial P_i}{\partial \delta_m} \end{bmatrix}$$

$N_{im} \quad H_{im}$

$\rightarrow$  Taking  $i$ th bus &  $m$ th bus are PV bus.

$$\begin{bmatrix} \vdots \\ \vdots \\ \frac{\partial P_i}{\partial \delta_m} \end{bmatrix}$$

$\rightarrow$  If  $i$ th PQ &  $m$ th bus is PV bus.

$$\begin{bmatrix} \vdots \\ \vdots \\ \frac{\partial Q}{\partial V_m} \quad \frac{\partial Q}{\partial \delta_m} \end{bmatrix}$$



→ Write causes of i, m, n buses

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PQ.

i, m, n  
Bus.

$$\begin{bmatrix} \Delta P_i \\ \Delta Q_i \end{bmatrix} = \begin{bmatrix} \frac{\partial P_i}{\partial \delta_m} & \frac{\partial P_i}{\partial V_m} & |V_m| \\ \frac{\partial Q_i}{\partial \delta_m} & \frac{\partial Q_i}{\partial V_m} & |V_m| \end{bmatrix}$$



# ④ Derivation of Jacobian Elements

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Let  $Y_{ik} = G_{ik} + jB_{ik}$ .

$V_i = |V_i| \angle \delta_i$

$V_k = |V_k| \angle \delta_k$

$S_i = V_i \left[ \sum_{k=1}^n Y_{ik} V_k \right]^* \star$

$S_i = P_i + jQ_i = \sum_{k=1}^n |V_i| |V_k| (G_{ik} - jB_{ik}) \angle (\delta_i - \delta_k)$

$P_i = \sum_{k=1}^n |V_i| |V_k|$

$S_i = P_i + jQ_i = \sum |V_i| |V_k| (G_{ik} - jB_{ik}) \angle \cos(\delta_i - \delta_k) + j \sin(\delta_i - \delta_k) \}$

$P_i = \text{Re} \{ S_i \} = \sum_{k=1}^n |V_i| |V_k| \{ G_{ik} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k) \} \text{--- (A)}$

$Q_i = \text{Im} \{ S_i \} = \sum_{k=1}^n |V_i| |V_k|$

$\{ G_{ik} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k) \} \text{--- (B)}$



$$H_{ii} = \frac{\delta P}{\delta \delta}$$

Diagonal element of H matrix

$$H_{ii} = \frac{\delta P_i}{\delta \delta_i} = \sum_{k=1}^n |V_i| |V_k|$$

$$\left\{ -G_{ik} \sin(\delta_i - \delta_k) + B_{ik} \cos(\delta_i - \delta_k) \right\} \quad \text{--- (E)}$$

$$P_i = \operatorname{Re} \{ S_i^* \} = \sum_{k=1}^n |V_i| |V_k|$$

$$\left\{ G_{ik} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k) \right\} \quad \text{--- (A)}$$

$$P_i = G_{ii} |V_i|^2 + \sum_{k=1}^n |V_i| |V_k| \times$$

$$\left\{ G_{ik} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k) \right\} \quad \text{--- (C)}$$

$$Q_i = -B_{ii} |V_i|^2 + \sum_{k=1}^n |V_i| |V_k|$$

$$\left\{ G_{ik} \sin(\delta_i - \delta_k) - B_{ik} \cos(\delta_i - \delta_k) \right\} \quad \text{--- (D)}$$

Adding (D) & (C)

$$H_{ii} + Q_i = -B_{ii} |V_i|^2$$

$$\boxed{H_{ii} = -Q_i - B_{ii} |V_i|^2}$$

$$H_{ik} = \frac{\delta P_i}{\delta \delta_k} = |V_i| |V_k| \left\{ G_{ik} \sin(\delta_i - \delta_k) - B_{ik} \cos(\delta_i - \delta_k) \right\}$$



$$\frac{S_{P_2}}{S_{P_3}} = |V_2| |V_3| \{ G_{23} \sin(\delta_2 - \delta_3) - B_{23} \cos(\delta_2 - \delta_3) \}$$

(E)

(A)